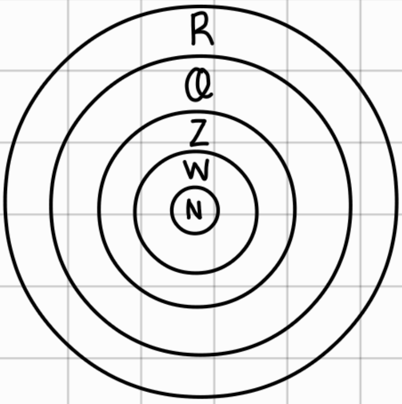
bilangan



N = bil. asli $\{1, 2, 3, \ldots\}$

W = bil. cacah $\{0, 1, 2, 3 \ldots\}$

Z = bil. bulat $\{-2, -1, 0, 1, 2, \ldots\}$

Q = bil. rasional $\{ q = \frac{a}{b} ,\ a, b \in Z,\ b \neq 0 \}$

R = bil. real (termasuk bil irasional)

I = bil irasional $\{ \pi, \sqrt{2}, \sqrt{7}$, 1,101001000 (bil. desimal tdk berpola) $\}$

bil. genap $\{2, 4, 6, 8 \ldots\}$

habis dibagi 2

bil. ganjil $\{1, 3, 5, 7 \ldots\}$

tdk habis dibagi 2

bil. prima $\{2, 3, 5, 7, 11, 13$, dst.$\}$

bil. yg hanya memiliki 2 faktor < angka 1 / angka itu sendiri

bil. komposit

bil lebih dari 1, selain prima

bil. kuadrat $\{4, 9, 16, 25$, dst.$\}$

bil hasil perkalian 2 bil. yg sama

satuan bilangan berpangkat

berulang tiap 1x = 5, 6

berulang tiap 2x = 4, 9

berulang tiap 4x = 2, 3, 7, 8

cth : $4^1 = 4$, $4^2 = 16$, $4^3 = 64$, $4^4 = 256$ — berulang tiap 2x

cth: tentukan satuan dari $2^{20} \times 4^7$

$\frac{20}{4} = 5$ sisa 0 → ambil urutan ke 4 ($2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$)

$\frac{7}{2} = 3$ sisa 1 → urutan ke 1 ($4^1 = 4$, $4^2 = 16$) ; $6 \times 4 = 24$

KPK

cth KPK dari 7 dan 5

Kelipatan 7 = 7, 14, 28, 35

Kelipatan 5 = 5, 10, 15, 20, 25, 35

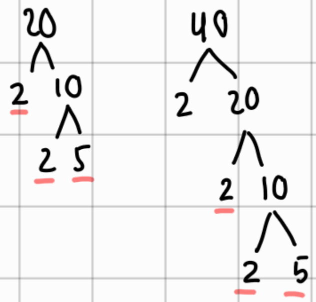
lampu A nyala tiap 7 detik, lampu B tiap 5, di detik brp mereka nyala bareng?

↳ KPK 7 dan 5 = 35

$\boxed{\text{note : KPK = menyamakan / barengan}}$

FPB

cth FPB dari 20 dan 40



$= 2^2 \times 5 = 20$

20 kue dan 40 permen dpt dibagi ke brp org agar dpt sama byk?

↳ FBP 40 20 = 20 org

$\boxed{\text{note : FPB = membagi rata tanpa sisa}}$

fungsi kuadrat

$y = ax^2 + bx + c$

$a \neq 0$

definit

positif	$= a > 0 \,,\ D < 0$
negatif	$: a < 0 \,.\ D < 0$

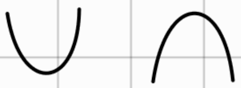
titik puncak /max /min /ekstrim / balik

$x_p = f'(x) = \dfrac{-b}{2a}$

$y_p = f(x_p) = \dfrac{D}{-4a}$

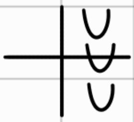
grafik fungsi kuadrat

• a

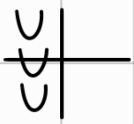


$a > 0$ $\quad$ $a < 0$

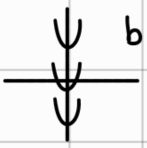
• b



kanan beda (kabe)
→ grafik di kanan sb y,
tanda b beda dgn a



kiri sama (kisa)
→ grafik di kiri sb y
tanda b sama dgn a

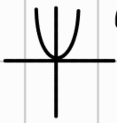


$b = 0$ grafik di tengah sumbu y

• c

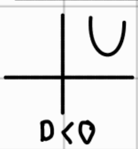
$c > 0$

$c < 0$

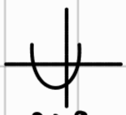


$c = 0$

• Diskriminan (D)



$D < 0$
tdk motong
tdk nyinggung



$D > 0$
memotong sb x di 2 titik

$D = 0$
menyinggung sb x

teorema vieta akar² fungsi kuadrat

$x_1 + x_2 = -\dfrac{b}{a}$

$x_1 \cdot x_2 = \dfrac{c}{a}$

$x_1 - x_2 = \pm \dfrac{\sqrt{D}}{a}$

operasi akar²

$x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2$

$x_1^3 + x_2^3 = (x_1 + x_2)^3 - 3x_1x_2(x_1x_2)$

$\dfrac{1}{x_1} + \dfrac{1}{x_2} = \dfrac{x_1 + x_2}{x_1x_2}$

menentukan persamaan

1) jika dik. titik puncak (x_p, y_p) dan titik yg dilewati (x, y)

$\boxed{y - y_p = a(x - x_p)^2}$ subs semua, cari a, stlh dpt a, subs x_p y_p dan a, x dan y ga usah

2) jika dik. tipot sb x $(x_1, 0), (x_2, 0)$ dan (x, y)

$\boxed{y = a(x - x_1)(x - x_2)}$ subs semua, cari a, stlh dpt a, subs $x_{1,2}$ $y_{1,2}$ dan a, x dan y ga usah

3) jika dik 3 titik yg dilewati

$\boxed{y = ax^2 + bx + c}$ (SPLTV)

sifat akar²

2 akar positif : $x_1 + x_2 > 0 ;\ D \geq 0$

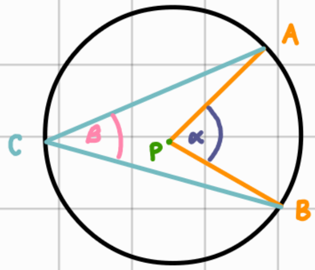
2 akar negatif : $x_1 + x_2 < 0 ;\ x_1x_2 > 0 ;\ D \geq 0$

saling berlawanan : $x_1 x_2 < 0 ; D > 0$

saling berkebalikan $x_1 x_2 = 1 \ ;\ D > 0$

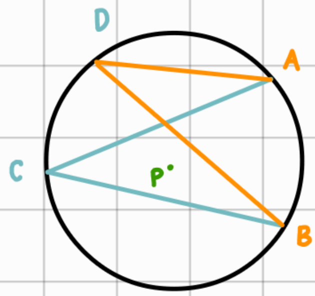
sudut

lingkaran



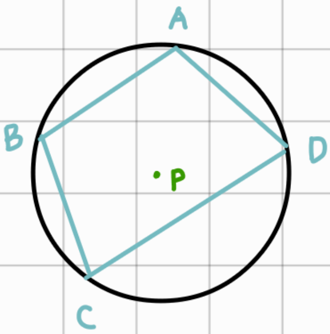
sudut pusat dan sudut keliling menghadap busur yg sama

$\boxed{\text{sudut pusat} = 2 \cdot \text{sudut keliling}}$



2 sudut keliling menghadap busur yg sama

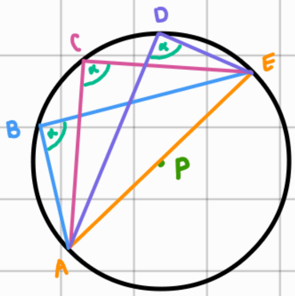
$\boxed{\angle ADB = \angle ACB}$



$\boxed{\text{jumlah 2 sudut yg berhadapan} = 180}$

$\angle ABC + \angle ADC = 180°$

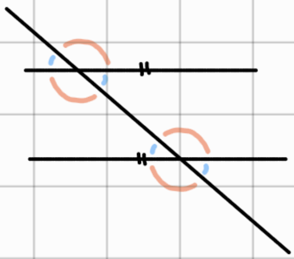
$\angle BAD + \angle BCD = 180°$



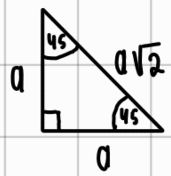
$\boxed{\text{sudut keliling yang menghadap diameter} \angle 90°}$

$\angle ABE = \angle ACE = \angle ADE = 90°$

garis

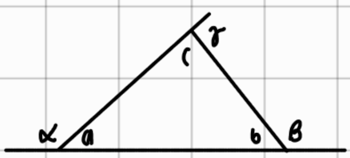


segitiga siku" sama kaki



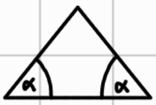
total sudut = 180°

luar segitiga



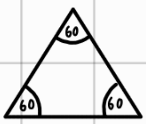
α	$= b + c$
β	$= a + c$
γ	$= a + b$

segitiga sama kaki



total sudut = 180°

segitiga sama sisi



Persamaan lingkaran

bentuk $\rightarrow (x-a)^2 + (y-b)^2 = r^2$ pusat (a,b)

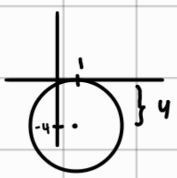
$$x^2+y^2 \underbrace{-2ax}_{A} \underbrace{-2by}_{B} + \underbrace{a^2+b^2-r^2}_{C} = 0$$

rms mencari pusat $= \boxed{-\frac{1}{2}A , -\frac{1}{2}B}$

rms mencari jari² $= \boxed{\sqrt{\underbrace{a^2+b^2}_{\text{dr pusat}} - \underbrace{C}_{\text{dr persamaan}}}} = \boxed{\sqrt{\underbrace{(-\frac{1}{2}A)^2 + (-\frac{1}{2}B)^2 - C}_{\text{dr persamaan}}}}$

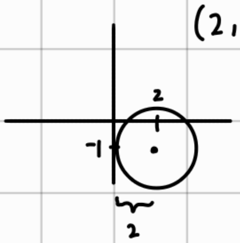
lingkaran dg pusat (a,b) menyinggung sb x $\rightarrow \boxed{r = |b|}$

$(1, -4)$



lingkaran dg pusat (a,b) menyinggung sb y $\rightarrow \boxed{r = |a|}$

$(2, -1)$



Kedudukan titik (x,y)

1) titik dalam lingkaran

   $(x-a)^2 + (y-b)^2 < r^2$

2) titik pada lingkaran

   $(x-a)^2 + (y-b)^2 = r^2$

3) titik luar lingkaran

   $(x-a)^2 + (y-b)^2 > r^2$

↳ subs x,y ke pers

* $(a,b) \rightarrow$ pusat ⊙

Kedudukan garis

1) memotong lingkaran di 2 titik berbeda

   $D > 0$

2) menyinggung lingkaran

   $D = 0$

3) tdk keduanya

   $D < 0$

↳ • garis $ax + by + c = 0$ diubah bentuk $y = \dots$

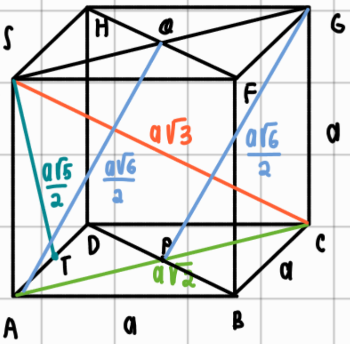
- subs y ke pers. lingkaran.
- cari D.

bangun ruang

kubus

L. permukaan : $6s$

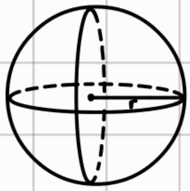
volume : s^3



bola

L. permukaan : $4\pi r^2$

volume : $\frac{4}{3}\pi r^3$

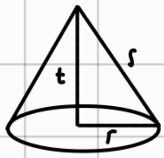


kerucut

L. permukaan : $\pi r\ (s + r)$

volume : $\frac{1}{3}\pi r^2 t$

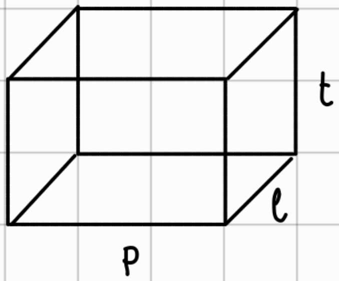
L. selimut : $\pi r s$



balok

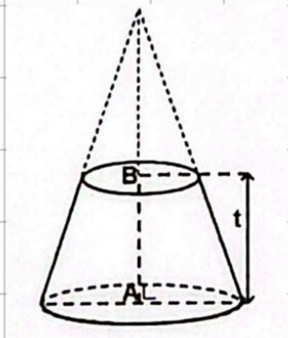
L. permukaan : $2(pl + lt + pt)$

volume : $p \cdot l \cdot t$



kerucut potong

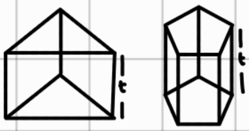
$$V_{potong} = \frac{1}{3}t\ (L_A + L_B + \sqrt{L_A \cdot L_B})$$



prisma

L. permukaan : $2L.a + Kel.a \cdot t$

volume : $L.a \cdot t$

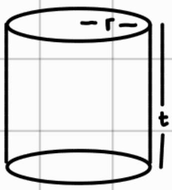


tabung

L. permukaan : $2\pi r\,(r + t)$

volume : $\pi r^2 t$

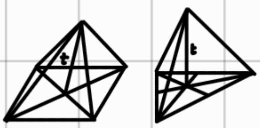
L. selimut : $2\pi r t$



L. permukaan tabung tanpa tutup : $\pi r\,(r + 2t)$

limas

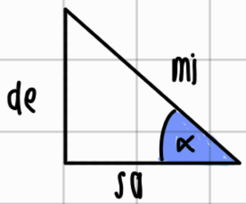
L. permukaan : L.a + jml. L. sisi



volume : $\frac{1}{3}$ L.a $\cdot$ t

L. selimut : byk $\triangle$ x L. $\triangle$

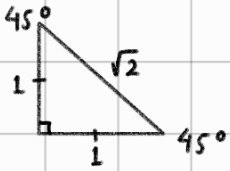
trigonometri



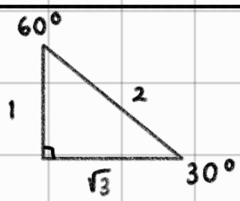
1. $\sin \alpha = \dfrac{de}{mi}$
2. $\cos \alpha = \dfrac{sa}{mi}$
3. $\tan \alpha = \dfrac{de}{sa}$
4. $\cot an\, \alpha = \dfrac{sa}{de}$
5. $\sec \alpha = \dfrac{mi}{sa}$
6. $\operatorname{cosec} \alpha = \dfrac{mi}{de}$

Perbandingan

* sudut 45°



* sudut 30 dan 60



sudut negatif (searah jarum jam)

$\sin(-\alpha) = -\sin \alpha$
$\cos(-\alpha) = \cos \alpha$
$\tan(-\alpha) = -\tan \alpha$

tabel

	0°	30°	37°	45°	53°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{1}{2}\sqrt{2}$	$\frac{4}{5}$	$\frac{1}{2}\sqrt{3}$	1
cos	1	$\frac{1}{2}\sqrt{3}$	$\frac{4}{5}$	$\frac{1}{2}\sqrt{2}$	$\frac{3}{5}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{3}\sqrt{3}$	$\frac{3}{4}$	1	$\frac{4}{3}$	$\sqrt{3}$	∞

Kuadran

Kuadran II (sin, cosec +) 90° — 180° $(180 - \alpha)$	Kuadran I (semuanya +) 0° — 90° α
Kuadran III (tan, cotan +) 180° — 270° $(180° + \alpha)$	Kuadran IV (cos, secan) 270° — 360° $(360 - \alpha)$

persamaan garis

menentukan persamaan

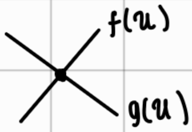
$\boxed{y - y_1 = m(x - x_1)}$ dik (x_1, y_1) dan m

$\boxed{m = \dfrac{y_2 - y_1}{x_2 - x_1}}$ dit → m
dik (x_1, y_1) dan (x_2, y_2)

$\boxed{\dfrac{y - y_1}{y_2 - y_1} = \dfrac{x - x_1}{x_2 - x_1}}$ dit → pers. grs
dik → melalui (x_1, y_1) dan (x_2, y_2)

penyelesaian

1 penyelesaian → garis berpotongan di 1 titik



eliminasi & subs

tdk punya penyelesaian / solusi → sejajar

$m_1 = m_2$

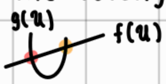
$\boxed{\dfrac{a}{p} = \dfrac{b}{q} \neq \dfrac{c}{r}}$

punya byk (∞) penyelesaian → berhimpit

$ax + by = c$
$px + qy = r$

$\boxed{\dfrac{a}{p} = \dfrac{b}{q} = \dfrac{c}{r}}$

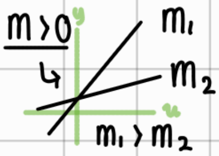
* garis f(x) memotong grafik g(x) pd 2 titik



↳ f(x) = g(x) akan membentuk pers. kuadrat, akar"nya adalah titik x, subs x ke garis, dpt titik y → (x_1, y_1) dan (x_2, y_2)

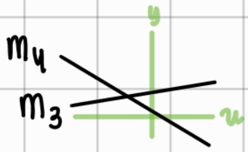
gradien

m > 0



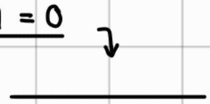
↳ $m_1 > m_2$

m < 0



↳ $m_3 > m_4$

m = 0



tdk punya m

gradien garis singgung

$$\boxed{\begin{aligned} m &= f'(x) \\ &= f'(x_1) \end{aligned}}$$
↳ subs x dik.

garis sejajar → $\boxed{m_1 = m_2}$

garis tegak lurus → $\boxed{m_1 = -\dfrac{1}{m_2}}$

jarak antar titik

(x_1, y_1) & (x_2, y_2)

$\boxed{r = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}}$

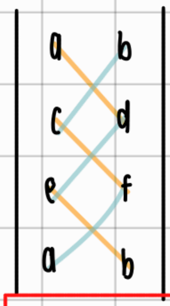
jarak titik ke garis

titik (a, b) garis y = px + qy + s

$\boxed{r = \left| \dfrac{p(a) + q(b) + s}{\sqrt{p^2 + q^2}} \right|}$

luas segitiga tiga titik koordinat

(a, b) (c, d) (e, f)



× $\frac{1}{2}$

$\boxed{|ad + cf + eb - bc - de - fa| \times \frac{1}{2}}$

bangun datar

	simetri lipat	simetri putar	luas
persegi	4	4	s^2
persegi panjang	2	2	$p\ell$
belah ketupat	2	2	$\frac{1}{2} d_1 \cdot d_2$
segitiga sama sisi	3	3	$\frac{1}{2} a \cdot t$
segitiga sama kaki	1	1	
segitiga siku-siku	0	1	
layang - layang	1	1	$\frac{1}{2} d_1 \cdot d_2$
jajargenjang	0	2	$a \cdot t$
trapesium sembarang	0	1	$\frac{\text{jml. sisi sejajar} \cdot t}{2}$
trapesium siku-siku	0	1	
trapesium sama kaki	1	1	
lingkaran	∞	∞	πr^2

aritmatika sosial

% untung & % rugi

$$\% \text{ untung} = \frac{\text{untung}}{\text{beli}} \cdot 100\% = \frac{(\text{jual} - \text{beli})}{\text{beli}} \cdot 100\%$$

$$\% \text{ rugi} = \frac{\text{rugi}}{\text{beli}} \cdot 100\% = \frac{(\text{beli} - \text{jual})}{\text{beli}} \; 100\%$$

diskon

$$\text{harga setelah diskon} = (1 - a\%) \cdot \text{harga awal}$$

aritmatika

suku ke-n

$$U_n = a + (n-1)b$$

beda

$$b = U_n - U_{(n-1)}$$

$$b = \frac{U_x - U_y}{x - y}, \quad x > y$$

beda baru

$$b' = \frac{b}{k+1}$$

k: byk sisipan

suku tengah

$$U_t = \frac{a + U_n}{2}$$

jumlah n suku pertama

$$S_n = \frac{n}{2}(2a + (n-1)b)$$

$$S_n = \frac{n}{2}(a + U_n)$$

aritmatika bertingkat

a: 2   5   10   17

b: 3   5   7

c: 2   2

$$U_n = a + (n-1)b + \frac{(n-1)(n-2)c}{2}$$

geometri

suku ke-n

$$U_n = ar^{n-1}$$

rasio

$$r = \frac{U_n}{U_{n-1}}$$

jumlah n suku pertama

$r > 1$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$r < 1$

$$S_n = \frac{a(1 - r^n)}{1 - r}$$

deret tak hingga

$$S_\infty = \frac{a}{1 - r}, \quad -1 < r < 1$$

suku tengah

$$U_t = \sqrt{U_{kanan} \times U_{kiri}}$$

geometri bertingkat

A: 200   120   80   60   50

a: −80   −40   −20   −10

r: $\times \frac{1}{2}$ $\times \frac{1}{2}$

$$U_n = A + S_{n-1}$$

$r < 1$ ↳ $U_n = A + \frac{a(1 - r^{n-1})}{1 - r}$

$r > 1$ ↳ $U_n = A + \frac{a(r^{n-1} - 1)}{r - 1}$

Statistika

mean (rata²)

$$\bar{x} = \frac{\text{jumlah nilai}}{\text{byk nilai}}$$

rata" gabungan

$$\bar{x}_{gab} = \frac{\bar{x}_1 n_1 + x_2 n_2}{n_1 + n_2}$$

quartil

Q_1 = Q bawah

Q_2 = Q tgh (median)

Q_3 = Q atas

median (nilai tgh)

data ganjil → n data , $me = x_{\frac{(n+1)}{2}}$

data genap → n data , $me = \dfrac{x_{\frac{n}{2}} + x_{\frac{(n+1)}{2}}}{2}$

simpangan quartil

$$sq = \frac{1}{2}(Q_3 - Q_1)$$

modus → data dgn frekuensi terbanyak (paling sering muncul)

* dik soal data tdk memiliki modus : frekuensi tiap data sama

varian / ragam

$$var = \frac{\Sigma(\bar{x} - x_i)^2}{n}$$

simpangan baku

$$sb = \sqrt{ragam}$$

data dikatrol

1) mean, median, modus, quartil
   +, −, ×, ÷ → berubah

2) jangkauan, simpangan

   +, − : tetap

   ×, ÷ : berubah

jangkauan

$$j : x_{max} - x_{min}$$

habis di bagi

habis dibagi 3

jumlah semua digit habis dibagi 3

$\rightarrow$ 2418 = 2+4+1+8 = 15 ✓

habis dibagi 4

dua digit terakhir habis dibagi 4

563<u>24</u> ✓

habis dibagi 6

- bilangan genap
- jumlah semua digit habis dibagi 3

habis dibagi 7

kalikan digit satuan dgn 2, hasilnya menjadi pengurang sisa digitnya, ulangi hingga sisa 2 digit, sisa 2 digit tsb habis di bagi 7

43778 = 4377 − 16 = 4361

436 − 2 = 434

43 − 8 = 35 ✓

habis dibagi 8

tiga digit terakhir habis di bagi 8

habis dibagi 9

jumlah semua digit habis dibagi 9

habis dibagi 11

- cara 1 $\rightarrow$ 9372 = 9 − 3 + 7 − 2 = 11

  528 = 5 − 2 + 8 = 11

- cara 2 $\rightarrow$ pisahkan dgn digit satuannya, sisa digitnya dikurangi dgn digit satuannya, hasilnya hbs dibagi 11

  2838 = 283 − 8 = 275

  27 − 5 = 22 ✓

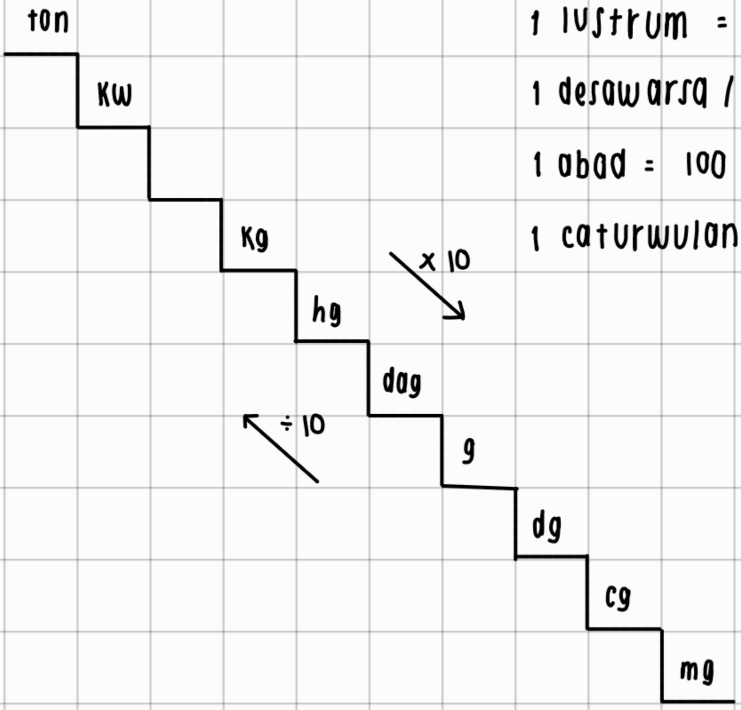
  5379 = 537 − 9 = 528

  52 − 8 = 44 ✓

Konversi satuan

massa

1 pon = 0,5 kg



waktu

1 windu = 8 tahun

1 lustrum = 5 tahun

1 desawarsa / dekade = 10 tahun

1 abad = 100 thn

1 caturwulan = 4 bulan

pcs

1 lusin = 12 pcs

1 kodi = 20 pcs

1 gross = 12 lusin = 144 pcs

1 rim = 500 pcs

volume

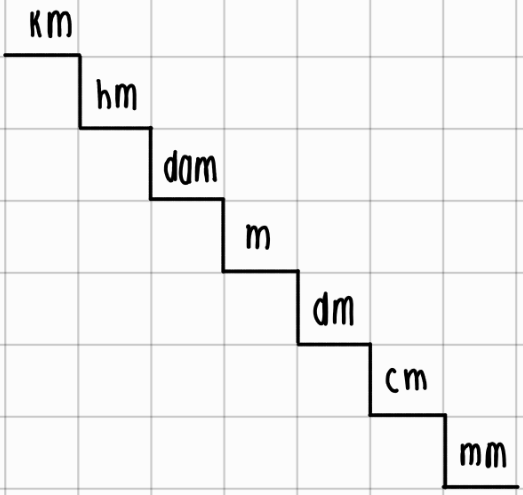
$1\ \ell = 1\ dm^3$

$1\ m\ell = 1\ cm^3$

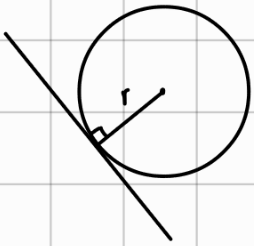
$1\ cc = 1\ cm^3$

$1\ \ell = 1000\ m\ell = 1000\ cm^3$

jarak



garis singgung lingkaran

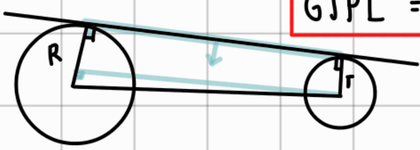


garis singgung persekutuan dalam

$$GSPD = \sqrt{(\text{jarak pusat})^2 - (R+r)^2}$$

garis singgung persekutuan luar

$$GSPL = \sqrt{(\text{jarak pusat})^2 - (R-r)^2}$$



turunan

$f(u) = au^n + bu^m + \dots$

$$f'(u) = an\,u^{(n-1)} + mb\,u^{(m-1)}$$

bentuk perkalian

$f(u) = uv$

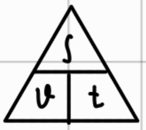
$$f'(u) = u'v + v'u$$

bentuk pembagian

$f(u) = \dfrac{u}{v}$

$$f(u) = \frac{u'v - v'u}{v^2}$$

jarak, kec. waktu



$$s = v \cdot t$$

"telat u menit" $\rightarrow$ $t + u$

"u menit sebelum" $\rightarrow$ $t - u$

bunga

bunga tunggal

$$Mn = Mo(1 + n \cdot i\%)$$

$$B = Mo \cdot i \cdot n$$

bunga majemuk

$$Mn = Mo(1 + i\%)^n$$

peluang

peluang

$$P = \frac{n(A)}{n(S)}$$

peluang tak saling lepas

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

peluang kejadian saling lepas

$$P(A \cup B) = P(A) + P(B)$$

peluang saling bebas

$$P(A \cap B) = P(A) \cdot P(B)$$

kombinasi

$$nCr = \frac{n!}{(n-r)!\,r!}$$

tanpa urutan

permutasi

$$nPr = \frac{n!}{(n-r)!}$$

dgn urutan

desimal berulang

2 angka $\rightarrow$ bagi 99

$0{,}242424\ldots \rightarrow \dfrac{24}{99}$

3 angka $\rightarrow$ bagi 999

$0{,}455455\ldots \rightarrow \dfrac{455}{999}$

transmisi dan gerak rotasi (gear, katrol, kipas)

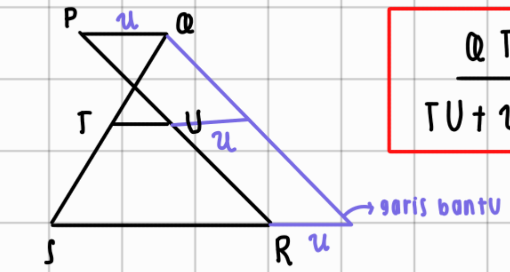
rasio putaran

$$n_1 r_1 = n_2 r_2$$

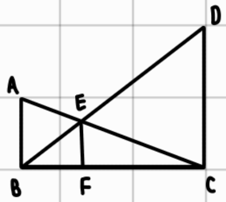
panjang lintasan

$$s = n \cdot 2\pi r$$

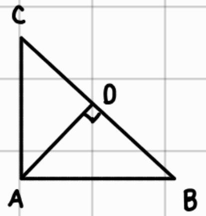
Kesebangunan



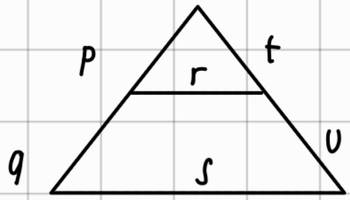
$$\frac{QT}{TU+u} = \frac{QS}{SR+u}$$



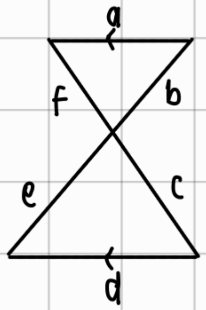
$$\frac{1}{AB} + \frac{1}{DC} = \frac{1}{EF}$$



$AD^2 = CD \times DB$
$AB^2 = BD \times BC$
$AC^2 = CD \times CB$



$$\frac{p}{p+q} = \frac{r}{s} = \frac{t}{t+u}$$



$$\frac{a}{d} = \frac{b}{e} = \frac{f}{c}$$

akar tak hingga

- $\sqrt{n\sqrt{n\sqrt{n..}}} = n$
- $\sqrt{n \div \sqrt{n \div \sqrt{n} \div ..}} = \sqrt[3]{n}$
- $\sqrt{n + \sqrt{n + \sqrt{n} + ..}} = p$

  $p \times q$ ambil yg besar
- $\sqrt{n - \sqrt{n - \sqrt{n} - ..}} = q$

  $p \times q$ ambil yg kecil
- $\sqrt{n \pm \sqrt{n \pm \sqrt{n} \pm ..}} = m$

  $n\,? \rightarrow \sqrt{n+m} = m \quad (\ )^2$

- $\sqrt{au + \sqrt{au + \sqrt{au}}\ ..} = b$

  $au = (b-1)\,b$

  ↳ $\sqrt{7u + \sqrt{7u + \sqrt{7u} + ..}} = 21$

  $\cancel{7}u = 20 \cdot \cancel{21}^{3}$

  $u = 60$
- $\sqrt{au - \sqrt{au - \sqrt{au}}\ ..} = b$

  $au = (b+1)\,b$

  ↳ $\sqrt{7u - \sqrt{7u - \sqrt{7u} - ..}} = 13$

  $\cancel{7}u = \cancel{14}^{2} \cdot 13$

  $u = 26$

transformasi geometri

translasi $\begin{pmatrix} a \\ b \end{pmatrix}$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$$

translasi garis $\begin{pmatrix} a \\ b \end{pmatrix}$

$$y - b = f(x - a)$$

refleksi

a) thd sb x $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} x \\ -y \end{pmatrix}$

b) thd sb y $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} -x \\ y \end{pmatrix}$

c) thd pusat $(0,0)$ $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} -x \\ -y \end{pmatrix}$

d) thd garis $y = x$ $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} y \\ x \end{pmatrix}$

e) thd garis $y = -x$ $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} -y \\ -x \end{pmatrix}$

f) thd garis $x = K$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} 2K - x \\ y \end{pmatrix}$

g) thd garis $y = K$

$\begin{pmatrix} x \\ y \end{pmatrix} \longrightarrow \begin{pmatrix} x \\ 2K - y \end{pmatrix}$

rotasi

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x - a \\ y - b \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$$

$(a,b) \rightarrow$ pusat rotasi

dilatasi

pusat (0,0)

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

pusat (a,b)

$$x' = a + K(x - a)$$

$$y' = b + K(y - b)$$

fungsi, komposisi, invers

daerah asal (domain)

1) $y = \dfrac{f(x)}{g(x)} \longrightarrow g(x) \neq 0$

2) $y = \sqrt{f(x)} \longrightarrow f(x) \geq 0$

daerah hasil (range)

1) $y = ax + b$

$-\infty < y < \infty$

2) $y = \sqrt{f(x)} \quad f(x) \geq 0$

sifat - sifat

1) $f \circ g(x) = f(g(x))$

2) $f \circ g(x) = h(x)$

$$\begin{cases} f(x) = (h \circ g^{-1})(x) \\ g(x) = (f^{-1} \circ h)(x) \end{cases}$$

3) $(f^{-1}(x))^{-1} = f(x)$

4) $f^{-1} \circ f(x) = f \circ f^{-1}(x) = f(x)$

5) $f^{-1}(a) = b \leftrightarrows f(b) = a$

6) $(f \circ g)^{-1}(x) = g^{-1} \circ f^{-1}(x)$

invers

1) $f(x) = ax + b$

$f^{-1}(x) = \dfrac{x - b}{a}$

2) $f(x) = \dfrac{ax + b}{cx + d}$

$f^{-1}(x) = \dfrac{-dx + b}{cx - a}$

10. Jika $f(x) = \frac{x-2011}{x-1}$, maka $f \circ f \circ f \circ f \circ f(x) = \ldots$

5 4 3 2 1

(A) $\dfrac{x+2011}{x-1}$

(B) $\dfrac{x-2011}{x+1}$

(C) $\dfrac{x-2011}{x+1}$

(D) $\dfrac{x-2011}{x-1}$

(E) $\dfrac{-x+2011}{x-1}$

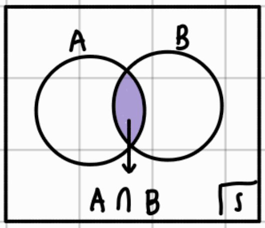
make sure $f^{-1}(x) = f(x)$

ganjil $= f(x)$

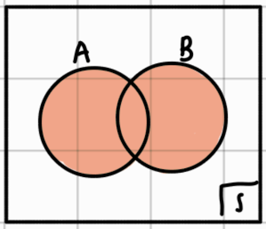
genap $= x$

himpunan

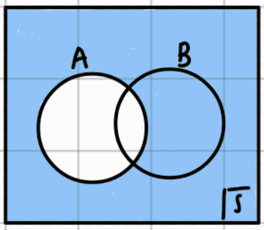
irisan (∩)



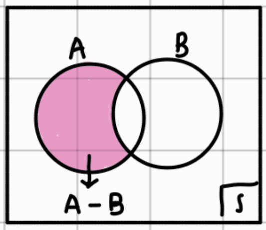
gabungan (∪)



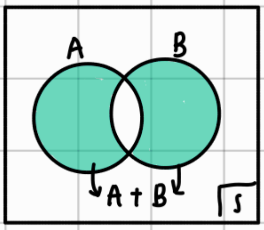
komplemen (A^c)



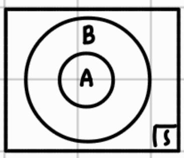
selisih (-)



jumlah (+)



bagian



$A \subseteq B$

eksponen

operasi

1) $a^m \cdot a^n = a^{m+n}$
2) $\dfrac{a^m}{a^n} = a^{m-n}$
3) $(a^m)^n = a^{m \cdot n}$
4) $(ab)^m = a^m \cdot b^m$
5) $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
6) $\left(\dfrac{a}{b}\right)^m = \dfrac{a^m}{b^m}$
7) $a^0 = 1$; $a \neq 0$
8) $a^{-m} = \dfrac{1}{a^m}$

merasionalkan bentuk akar

1) $\dfrac{K}{\sqrt{a}} = \dfrac{K}{\sqrt{a}} \cdot \dfrac{\sqrt{a}}{\sqrt{a}}$
2) $\dfrac{K}{\sqrt{a} \pm \sqrt{b}} = \dfrac{K}{\sqrt{a} \pm \sqrt{b}} \times \dfrac{\sqrt{a} \mp \sqrt{b}}{\sqrt{a} \mp \sqrt{b}}$
3) $\dfrac{K}{a \pm \sqrt{b}} = \dfrac{K}{a \pm \sqrt{b}} \times \dfrac{a \mp \sqrt{b}}{a \mp \sqrt{b}}$
4) $\sqrt{(a+b) \pm 2\sqrt{a \cdot b}} = \sqrt{a} \pm \sqrt{b}$

penjabaran

- $(a \pm b)^2 = a^2 + b^2 \pm 2ab$
- $a^2 - b^2 = (a+b)(a-b)$

  $a^9 - b^4 = (a^3 + b^2)(a^3 - b^2)$
- $a^2 + b^2 = (a+b)^2 - 2ab$
- $u^3 - b^3 = (u-b)(u^2 + bu + b^2)$
- $u^3 + a^3 = (u+a)(u^2 - au + a^2)$

matriks

operasi

1) penjumlahan dan pengurangan

$A_{(p\times q)} \pm B_{(p\times q)} = C_{(p\times q)}$ ordo hrs sama

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \pm \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a\pm e & c\pm f \\ b\pm g & d\pm h \end{pmatrix}$$

2) perkalian

$$A_{(m\times n)} \times B_{(n\times p)} = C_{(m\times p)}$$

sama

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

- $A^2 = A\cdot A$
- $A\cdot B\cdot C = (AB)C$
  $= A(BC)$

3) perkalian dgn konstanta

$$k\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$$

transpose

$$A_{(p\times q)} \longrightarrow A^T_{(q\times p)}$$

$$A_{(2\times 3)} = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \qquad A^T_{(3\times 2)} = \begin{pmatrix} a & d \\ b & e \\ c & f \end{pmatrix}$$

- sifat":

1) $(A^T)^T = A$
2) $(A\pm B)^T = A^T \pm B^T$
3) $(A\cdot B)^T = B^T\cdot A^T$

determinan

$$|A|_{(2\times 2)} = \left|\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right| = ad - bc$$

$$|A|_{(3\times 3)} = \left|\begin{array}{ccc|cc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \end{array}\right|$$

$$= aei + bfg + cdh - ceg - afh - bdi$$

- sifat"

1) $|A\cdot B| = |A|\cdot|B|$
2) $|k\cdot A_{(n\times n)}| = k^n|A|$
3) $|A^T| = A$
4) matriks singular $\rightarrow |A| = 0$
   (tdk pny invers)

invers

$$A_{(2\times 2)} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^{-1} = \frac{1}{|A|}\ \text{adj}\ (A)$$

$$\text{adj} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

- sifat"

1) $(A^{-1})^{-1} = A$
2) $(A\cdot B)^{-1} = B^{-1}A^{-1}$
3) $|A^{-1}| = \dfrac{1}{|A|}$
4) $A\cdot A^{-1} = I \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
5) $AX = B$ | $XA = B$
   $X = A^{-1}B$ | $X = BA^{-1}$